

$$\begin{array}{c}
\mathcal{K}_{0^+}^{\#1} \quad \mathcal{K}_{0^+}^{\#2} \\
\mathcal{K}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \gamma_1 k^2 & \frac{\gamma_2 k^2}{2} \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline \frac{\gamma_2 k^2}{2} & \gamma_1 k^2 \\ \hline \end{array} \quad \mathcal{K}_{1^-}^{\#1}{}_\alpha \quad \mathcal{K}_{1^-}^{\#2}{}_\alpha \\
\mathcal{K}_{1^-}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline \frac{5 k^2 \kappa_2^{(4)}}{6} & -\frac{1}{6} \sqrt{5} k^2 \kappa_2^{(4)} \\ \hline \end{array} \\
\mathcal{K}_{1^-}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline -\frac{1}{6} \sqrt{5} k^2 \kappa_2^{(4)} & \frac{k^2 \kappa_2^{(4)}}{6} \\ \hline \end{array} \quad \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline 0 \\ \hline \end{array} \quad \mathcal{K}_{3^-}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{3^-}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline -\frac{3 k^2 \kappa_2^{(4)}}{2} \\ \hline \end{array}
\end{array}$$

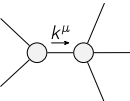
$$\begin{array}{c}
\mathcal{J}_{0^+}^{\#1} \quad \mathcal{J}_{0^+}^{\#2} \\
\mathcal{J}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \frac{\gamma_1 k^2}{\text{Det}(0^+)} & -\frac{\gamma_2 k^2}{2 \text{Det}(0^+)} \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline -\frac{\gamma_2 k^2}{2 \text{Det}(0^+)} & \frac{\gamma_1 k^2}{\text{Det}(0^+)} \\ \hline \end{array} \quad \mathcal{J}_{1^-}^{\#1}{}_\alpha \quad \mathcal{J}_{1^-}^{\#2}{}_\alpha \\
\mathcal{J}_{1^-}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline \frac{5}{6 k^2 \kappa_2^{(4)}} & -\frac{\sqrt{5}}{6 k^2 \kappa_2^{(4)}} \\ \hline \end{array} \\
\mathcal{J}_{1^-}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline -\frac{\sqrt{5}}{6 k^2 \kappa_2^{(4)}} & \frac{1}{6 k^2 \kappa_2^{(4)}} \\ \hline \end{array} \quad \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline 0 \\ \hline \end{array} \quad \mathcal{J}_{3^-}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{3^-}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline -\frac{2}{3 k^2 \kappa_2^{(4)}} \\ \hline \end{array}
\end{array}$$

Abbreviations used in matrices

$$\gamma_1 == \kappa_1^{(4)} + \kappa_2^{(4)} \ \&\& \ \gamma_2 == 2 \kappa_1^{(4)} - \kappa_2^{(4)} \ \&\& \ \text{Det}(0^+) == \frac{3}{4} k^4 \kappa_2^{(4)} (4 \kappa_1^{(4)} + \kappa_2^{(4)})$$

Lagrangian

$$\kappa_1^{(4)} \partial_\beta \mathcal{K}_\chi^\delta \partial^\alpha \mathcal{K}_\alpha^{\beta} + \kappa_2^{(4)} \partial_\chi \mathcal{K}_\beta^\delta \partial^\alpha \mathcal{K}_\alpha^{\beta} + \frac{9}{2} \kappa_2^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\beta \mathcal{K}_\beta^\delta - 3 \kappa_2^{(4)} \partial^\chi \mathcal{K}_\alpha^{\beta} \partial_\delta \mathcal{K}_\beta^\delta - \frac{3}{2} \kappa_2^{(4)} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{1^-}^{\#2\alpha} + \mathcal{J}_{1^-}^{\#1\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}_\beta^{\beta\chi} == \partial_\chi \partial^\alpha \mathcal{J}_\beta^{\alpha\beta}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$	5	$k^\chi (\partial_\epsilon \partial_\chi \partial^\beta \partial^\alpha \mathcal{J}_\delta^{\delta\epsilon} + 2 \partial_\epsilon \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}_\chi^{\delta\epsilon} - 3 \partial_\epsilon \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\beta\delta\epsilon} -$ $3 \partial_\epsilon \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\alpha\delta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial_\delta \partial_\chi \mathcal{J}^{\alpha\beta\delta} + \eta^{\alpha\beta} \partial_\phi \partial_\epsilon \partial_\delta \partial_\chi \mathcal{J}^{\delta\epsilon\phi} - \eta^{\alpha\beta} \partial_\phi \partial^\phi \partial_\epsilon \partial_\chi \mathcal{J}_\delta^{\delta\epsilon}) == 0$	
Total # constraint(s):	8		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{1}{\kappa_2^{(4)}}$
Resolved unitarity condition(s):		$\kappa_2^{(4)} > 0$	